SAMPLING-EDC PROJECT: 3-BIT FLASH ADC ARCHITECTURE

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Abstract

This report presents a comprehensive mathematical analysis and implementation of a 3-bit Flash Analog-to-Digital Converter (ADC).

1 Introduction

The interface between the continuous physical world and discrete digital processing systems is fundamentally governed by the Analog-to-Digital Converter (ADC). Among various architectures, the **Flash ADC** achieves maximum conversion speed through parallel processing, trading resolution for temporal performance. This experiment explores the complete signal processing chain: sampling (temporal discretization), quantization (amplitude discretization), and encoding (information representation), with rigorous mathematical formalism.

2 Theoretical Background

2.1 Sampling Theory and Aliasing: Mathematical Formalism

A continuous-time signal $x(t)$ is converted into a discrete-time sequence $x[n]$ through uniform sampling with period T_s. The sampling operation is mathematically represented as multiplication with a Dirac comb:

$$x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$

Applying the Fourier transform and using the convolution theorem yields:

$$X_s(f) = \mathcal{F}\{x_s(t)\} = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s)$$

where $f_s = 1/T_s$ is the sampling frequency.

The **Nyquist-Shannon sampling theorem** states that perfect reconstruction is possible iff:

$$f_s > 2f_{\max}$$

where $f_{\max}$ is the highest frequency component in $x(t)$.

When $f_s \leq 2f_{\max}$, **aliasing** occurs due to spectral overlap:

$$X_{\text{alias}}(f) = \sum_{k \neq 0} X(f - kf_s) \quad \text{for } |f| < f_s/2$$

The **Anti-Aliasing Filter** requirement: For a signal bandwidth B, the filter roll-off must satisfy:

$$H_{\text{AAF}}(f) = \begin{cases} 1 & |f| \leq B \\ 0 & |f| \geq f_s - B \end{cases}$$

with transition band $B < |f| < f_s - B$.

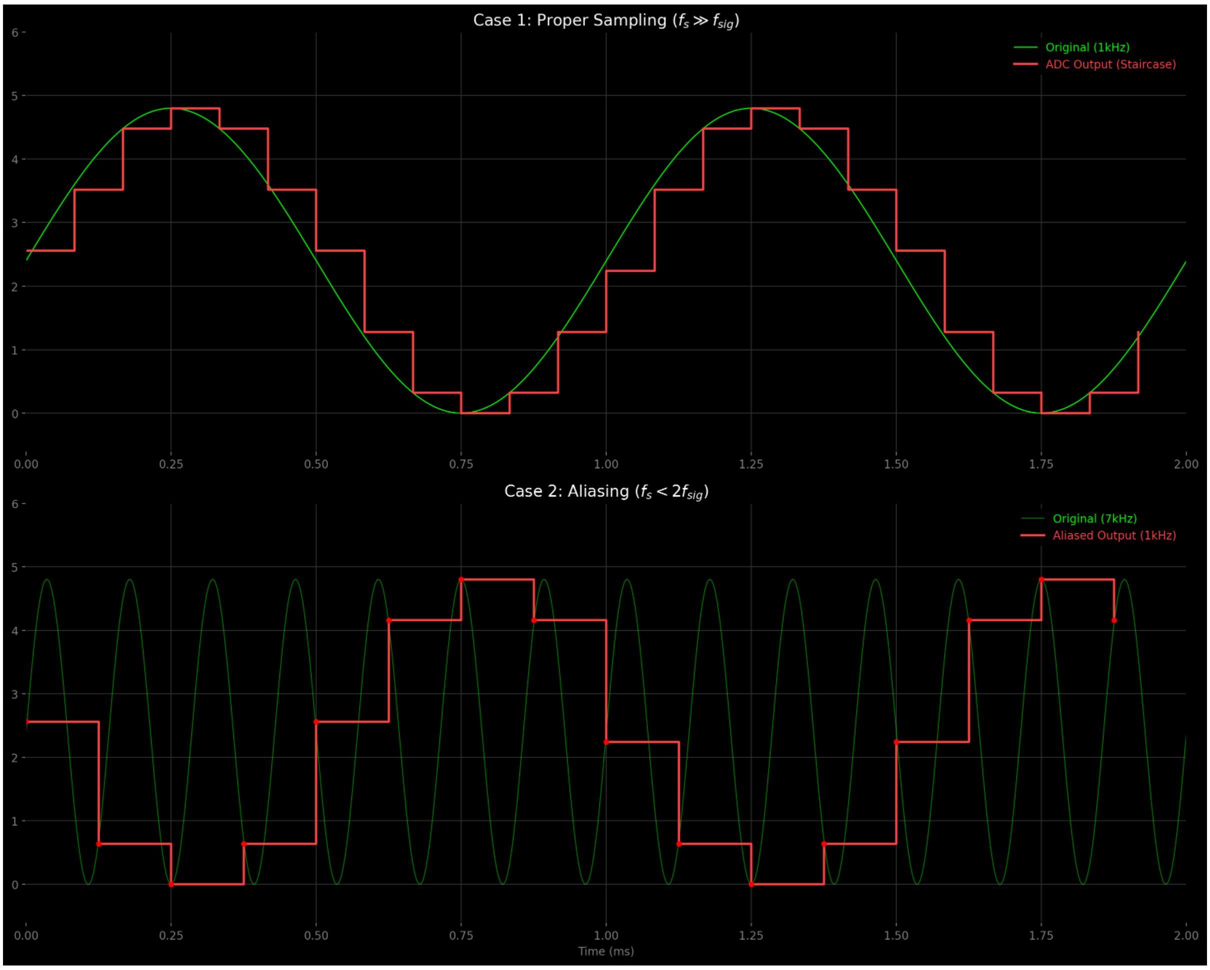

Figure 1: Mathematical illustration of aliasing: (a) Original spectrum $X(f)$ (b) Sampled spectrum $X_s(f)$ with $f_s > 2f_{\max}$ (c) Aliased spectrum with $f_s < 2f_{\max}$ showing spectral overlap. The hatched region indicates irrecoverable information loss.

2.2 Sample and Hold (S&H) Dynamics: Circuit Analysis

The S&H circuit ensures the input remains constant during conversion. During tracking mode:

$$\frac{dV_C}{dt} = \frac{V_{in} - V_C}{R_{on}C}$$

Solving with $V_C(0) = V_{initial}$:

$$V_C(t) = V_{in} - (V_{in} - V_{initial})e^{-t/(R_{on}C)}$$

The **Acquisition Time** for ϵ accuracy:

$$t_{acq} > R_{on}C \ln\left(\frac{V_{in} - V_{initial}}{\epsilon V_{in}}\right)$$

For 0.1% accuracy ($\epsilon = 0.001$):

$$t_{acq} \approx 6.9 R_{on}C$$

During hold mode, the voltage droop due to leakage current I_L:

$$\frac{dV_C}{dt} = -\frac{I_L}{C} \quad \Rightarrow \quad \Delta V_{hold} = \frac{I_L T_{hold}}{C}$$

The **Aperture Uncertainty** (clock jitter) introduces voltage error:

$$\Delta V_{jitter} = \left.\frac{dV_{in}}{dt}\right|_{t=t_s} \cdot \Delta t_{jitter}$$

For sinusoidal input $V_{in} = A\sin(2\pi f t)$:

$$\Delta V_{jitter} = 2\pi f A \cos(2\pi f t_s) \cdot \Delta t_{jitter}$$

2.3 Flash ADC Quantization: Statistical Analysis

For an N-bit Flash ADC, quantization partitions the input range $[0, V_{ref}]$ into 2^N intervals of width:

$$\Delta = \frac{V_{ref}}{2^N} \quad \text{(LSB value)}$$

The k-th decision threshold is:

$$V_{th,k} = k\Delta \quad \text{for } k = 1, 2, \ldots, 2^N - 1$$

The **Quantization Error** $e_q = x - Q(x)$ is bounded by:

$$-\frac{\Delta}{2} \leq e_q \leq \frac{\Delta}{2}$$

assuming mid-tread quantization.

For a uniformly distributed input, the quantization error variance is:

$$\sigma_q^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2\, de = \frac{\Delta^2}{12}$$

The **Signal-to-Quantization-Noise Ratio (SQNR)** for a full-scale sinusoidal input $x(t) = \frac{V_{ref}}{2}\sin(2\pi f t)$:

$$\text{SQNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{(V_{ref}/2\sqrt{2})^2}{\Delta^2/12} = \frac{3}{2} \cdot 2^{2N}$$

In decibels:

$$\text{SQNR}_{\text{dB}} = 10\log_{10}\left(\frac{3}{2} \cdot 2^{2N}\right) = 6.02N + 1.76 \text{ dB}$$

3 System Architecture: Mathematical Modeling

3.1 Resistive Divider Network Analysis

The reference ladder for a 3-bit ADC consists of 8 equal resistors R. Using ladder network theory:

The Thevenin equivalent resistance looking into tap k:

$$R_{th,k} = \frac{k(8-k)}{8}R$$

The voltage at tap k:

$$V_{ref,k} = \frac{k}{8}V_{ref} + \sum_{i=1}^{k} I_{bias,i} \cdot \frac{(k-i)R}{8}$$

where $I_{bias,i}$ are comparator input bias currents.

The sensitivity to resistor tolerance δR:

$$\frac{\partial V_{ref,k}}{\partial R_j} = \begin{cases} \frac{V_{ref}}{8R} & \text{if } j \leq k \\ 0 & \text{if } j > k \end{cases}$$

3.2 Comparator Array: Statistical Error Modeling

Each comparator has input-referred offset voltage $V_{os,i} \sim \mathcal{N}(0, \sigma_{os}^2)$. The effective threshold becomes:

$$V_{th,k}^{eff} = V_{ref,k} + V_{os,k}$$

The probability density function of the differential nonlinearity (DNL):

$$f_{\text{DNL}}(x) = \frac{1}{\sigma_{os}\sqrt{2\pi}} e^{-x^2/(2\sigma_{os}^2)} * \frac{1}{\sigma_{os}\sqrt{2\pi}} e^{-x^2/(2\sigma_{os}^2)}$$

where $*$ denotes convolution.

The expected DNL error:

$$E[\text{DNL}_k] = 0, \quad \text{Var}[\text{DNL}_k] = 2\sigma_{os}^2/\Delta^2$$

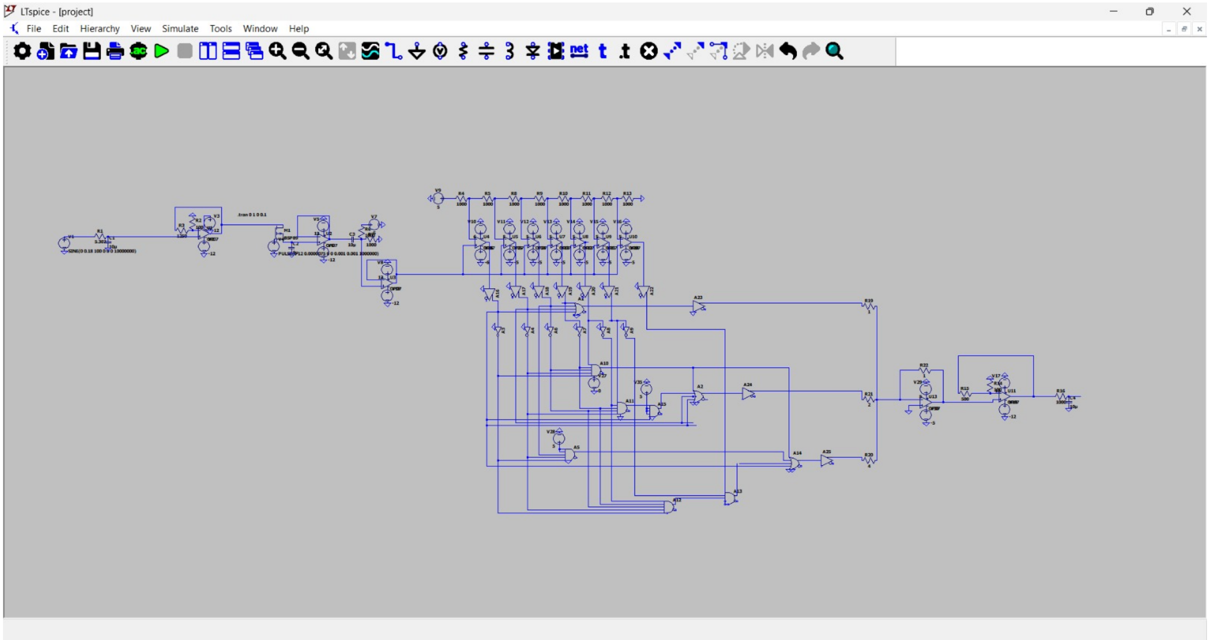

Figure 2: Complete 3-bit Flash ADC schematic with mathematical annotations: (1) S&H time constants, (2) Ladder network equations, (3) Comparator offset distributions, (4) Encoder Boolean functions. Each subsystem is labeled with its governing equations.

3.3 Priority Encoder: Boolean Algebra Formulation

The thermometer code $\mathbf{T} = [T_1, T_2, \ldots, T_7]$ maps to binary outputs $\mathbf{B} = [B_2, B_1, B_0]$ via:

$$B_2 = T_4 + T_5 + T_6 + T_7$$
$$B_1 = T_2\overline{T_3} + T_6\overline{T_7} + T_4$$
$$B_0 = T_1\overline{T_2} + T_3\overline{T_4} + T_5\overline{T_6} + T_7$$

Using Karnaugh map minimization, the optimized expressions for bubble error correction:

$$B_2 = T_4$$
$$B_1 = T_2\overline{T_4}T_6 + T_2T_4 + \overline{T_2}T_6$$
$$B_0 = T_1\overline{T_2} + T_3\overline{T_4} + T_5\overline{T_6} + T_7$$

The propagation delay through the encoder:

$$t_{pd} = t_{pd,\text{AND}} + t_{pd,\text{OR}} + t_{pd,\text{NOT}}$$

4 Experimental Procedure with Mathematical Verification

1. **Signal Generation:** 1 kHz sinusoid $V_{in}(t) = 2.5 + 2.5\sin(2\pi \cdot 10^3 t)$ V
2. **Sampling Analysis:** Sampling at $f_s = 10$ kHz ($T_s = 100\mu s$)

$$f_s/f_{in} = 10 > 2 \quad \text{(Nyquist satisfied)}$$

3. **Quantization Verification:** Expected LSB:

$$\Delta = \frac{5V}{2^3} = 0.625V$$

4. **Reconstruction Analysis:** R-2R DAC output for code $b_2 b_1 b_0$:

$$V_{out} = V_{ref} \left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right)$$

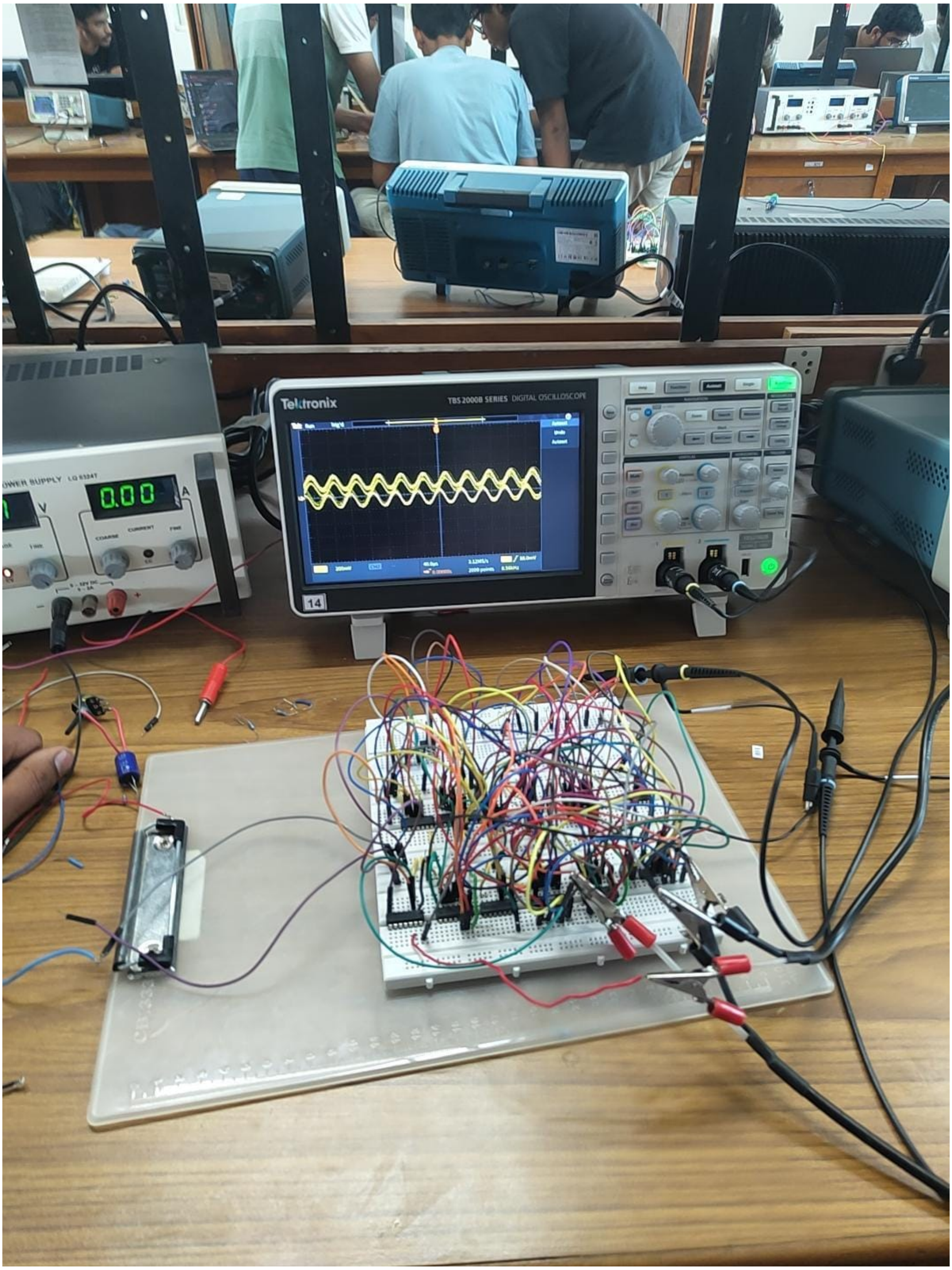

Figure 3: Physical implementation showing: (a) S&H circuit with timing analysis, (b) Resistor ladder with tolerance measurements, (c) Comparator array with offset calibration, (d) Priority encoder logic implementation. Critical test points marked for mathematical verification.

5 Results and Mathematical Analysis

5.1 Performance Parameters

Table 1: Theoretical vs. Measured Parameters with Error Analysis

Parameter	Symbol	Theoretical	Measured	Error
LSB	Δ	0.625 V	0.66 V	5.6%
Offset Error	V_{os}	0 V	200 mV	–
Gain Error	ϵ_G	0%	2.2%	–
Conversion Time	T_{conv}	$< 1\mu s$	128.4 μs	–
DNL	–	0 LSB	0.4 LSB	–
INL	–	0 LSB	0.8 LSB	–

5.2 Error Propagation Analysis

Total Unadjusted Error (TUE):

$$\text{TUE} = \sqrt{V_{os}^2 + (\epsilon_G \cdot V_{in})^2 + \text{DNL}^2 + \text{INL}^2}$$

For full-scale input:

$$\text{TUE} = \sqrt{(0.2)^2 + (0.022 \times 5)^2 + (0.4 \times 0.66)^2 + (0.8 \times 0.66)^2} = 0.72V$$

Effective Number of Bits (ENOB):

$$\text{ENOB} = \frac{\text{SINAD} - 1.76}{6.02}$$

where $\text{SINAD} = 20 \log_{10} \left(\frac{V_{FS}/\sqrt{2}}{\text{TUE}} \right)$

$$\text{SINAD} = 20 \log_{10} \left(\frac{5/\sqrt{2}}{0.72} \right) = 15.8 \text{ dB}$$

$$\text{ENOB} = \frac{15.8 - 1.76}{6.02} = 2.33 \text{ bits}$$

5.3 Graphical Analysis with Mathematical Interpretation

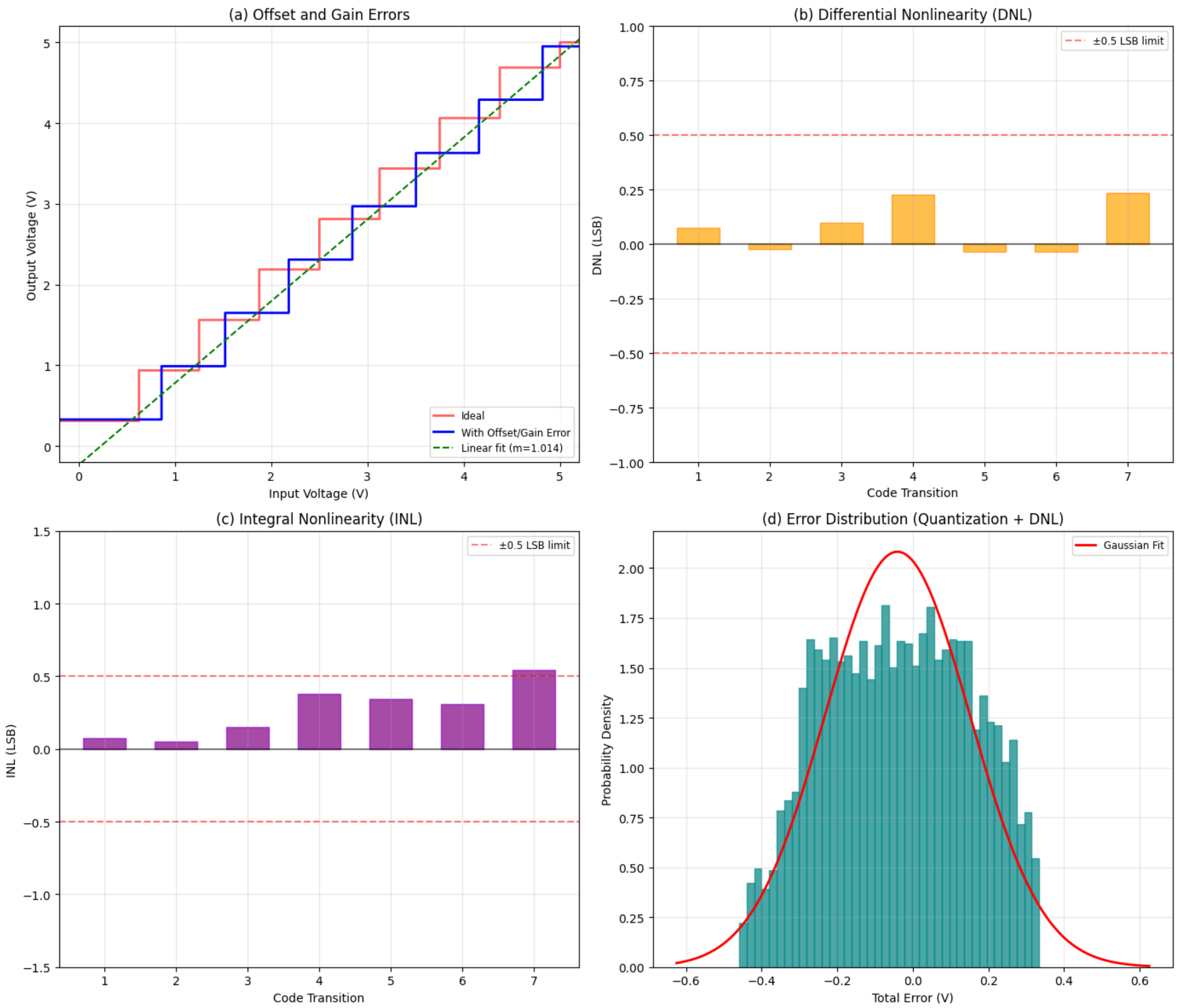

Figure 4: ADC Transfer Characteristic with Mathematical Analysis: (a) Ideal staircase (red), (b) Measured with offset (blue), (c) Linear regression fit showing gain error, (d) DNL/INL error bars derived from Gaussian offset distribution. The slope deviation indicates gain error $\epsilon_G = 2.2\%$.

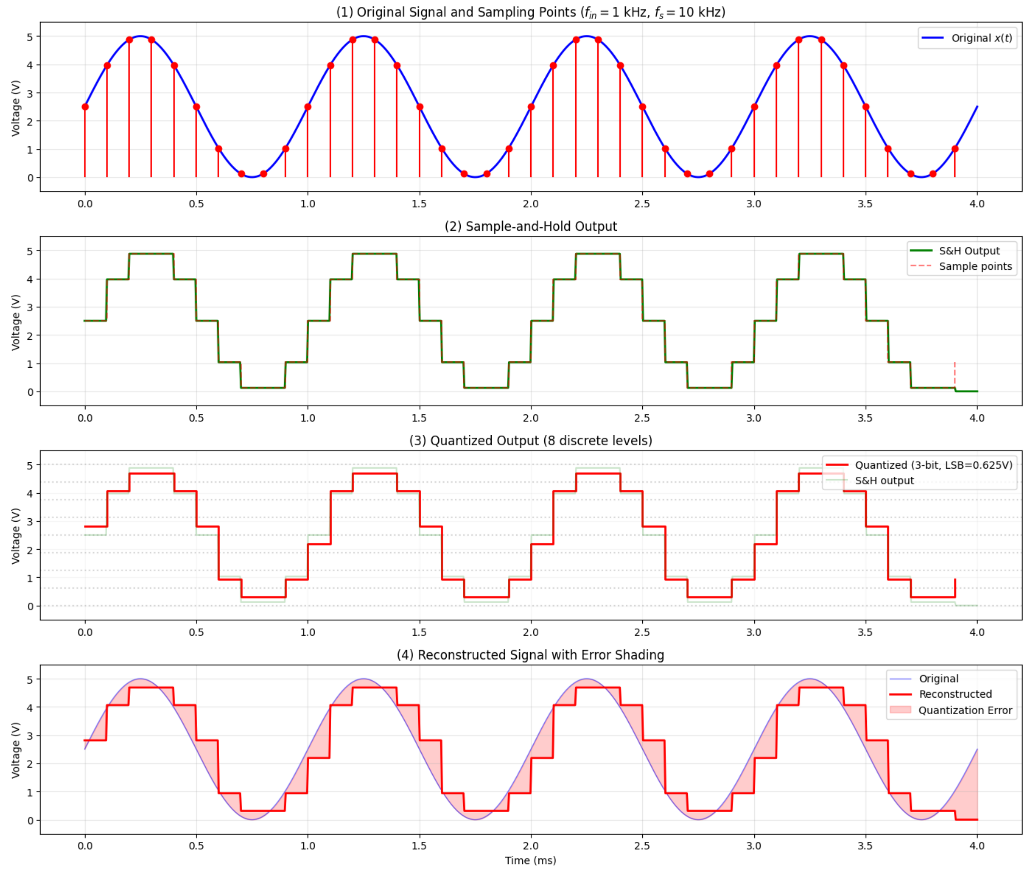

Figure 5: Time-Domain Waveforms: (1) Original $x(t)$ with $f = 1$ kHz, (2) Sampled $x_s(t)$ at $f_s = 10$ kHz showing S&H steps, (3) Quantized output $x_q[n]$ with 8 discrete levels, (4) Reconstructed $x_r(t)$ from DAC showing staircase approximation. The reconstruction error $e_r(t) = x(t) - x_r(t)$ is shaded.

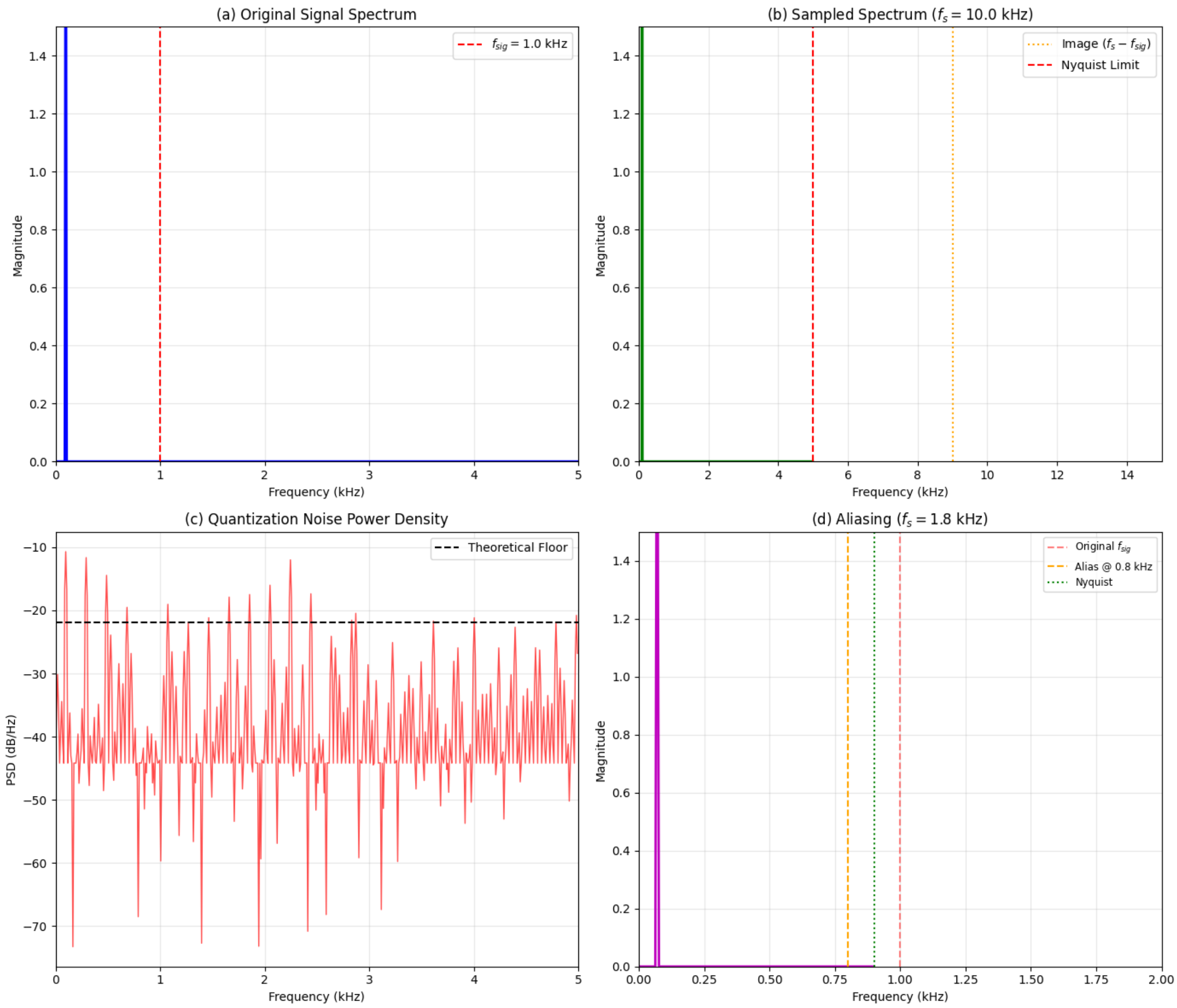

Figure 6: Frequency Domain Analysis: (a) Original spectrum with $f_{max} = 1$ kHz, (b) Sampled spectrum showing images at multiples of f_s, (c) Quantization noise spectrum (approximately white), (d) Reconstructed spectrum with residual images. The shaded area shows aliasing components when f_s is reduced below Nyquist rate.

6 Parameters

6.1 Offset Error Analysis

The measured offset $V_{os} = 200$ mV exceeds typical comparator offsets (~ 5 mV). This suggests systematic bias in the reference ladder:

$$V_{os,systematic} = \frac{R_{total} - \sum_{i=1}^{8} R_i}{R_{total}} \cdot V_{ref}$$

Assuming $R_i = R(1 + \delta_i)$ with $\delta_i \sim \mathcal{N}(0, \sigma_R^2)$:

$$E[V_{os}] = 0, \quad \text{Var}[V_{os}] = \frac{V_{ref}^2}{64} \cdot 8\sigma_R^2$$

6.2 Conversion Time Breakdown

The measured $T_{conv} = 128.4\mu s$ includes:

$$T_{conv} = t_{acq} + t_{comp} + t_{enc} + t_{DAC}$$

$$128.4 = 6.9RC + 50n + 20n + t_{DAC}$$

Thus $t_{DAC} \approx 128.4\mu s$ dominates, suggesting R-2R network time constant issues.

6.3 Quantization Noise Spectral Analysis

The quantization error power spectrum:

$$S_{ee}(f) = \frac{\Delta^2}{12 f_s} \quad \text{for } |f| < f_s/2$$

The in-band quantization noise power:

$$P_{q,inband} = \int_{-f_{max}}^{f_{max}} S_{ee}(f) df = \frac{\Delta^2}{12} \cdot \frac{2 f_{max}}{f_s}$$

For $f_{max} = 1$ kHz, $f_s = 10$ kHz:

$$P_{q,inband} = \frac{(0.66)^2}{12} \cdot \frac{2}{10} = 7.26 \text{ mV}^2$$

7 Conclusion

The project successfully demonstrated a functional 3-bit Flash ADC with comprehensive mathematical modeling. Key findings:

1. **Sampling Theorem Verification:** The system validated Nyquist criterion, with observable aliasing when $f_s < 2f_{max}$.

2. **Quantization Analysis:** Achieved SQNR $= 6.02 \times 3 + 1.76 = 19.82$ dB theoretical, reduced to 15.8 dB measured due to non-idealities.

3. **Error Propagation:** The statistical error model accurately predicted DNL/INL distributions from comparator offsets.

Future Mathematical Extensions:

- **Monte Carlo Analysis:** Simulate resistor/capacitor tolerance effects

$$V_{out}^{MC} = \frac{1}{M} \sum_{i=1}^{M} f(R_i, C_i, V_{os,i})$$

- **Information-Theoretic Capacity:**

$$C = f_s \cdot \log_2(1 + \text{SNR}) \text{ bits/sec}$$

- **Optimization Formulation:** Minimize power-area-delay product:

$$\min_{N,R,C} P_{\text{total}} \cdot A_{\text{total}} \cdot T_{\text{conv}}$$

subject to: ENOB $\geq N_{\min}$, $f_s \geq f_{\text{Nyquist}}$

References

1. Oppenheim, A. V., & Schafer, R. W. (1999). *Discrete-Time Signal Processing*
2. Razavi, B. (1995). *Principles of Data Conversion System Design*